Standards, FHIR & AI in Healthcare

Workshop UPNA 2026 – Day 2

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Agenda – Day 2

1. Recap of Day 1
2. Web app + FHIR demo
3. Standards and AI in healthcare (mini-module)
4. Group activity: design your own FHIR-based use case
5. Lightning presentations & wrap-up

Day 1 Recap

- FHIR basics: resources, references, JSON representation
- RESTful operations: GET, POST, PUT, DELETE
- Lab 1: creating Patients and Observations
- Lab 2: mini workflow with DiagnosticReport

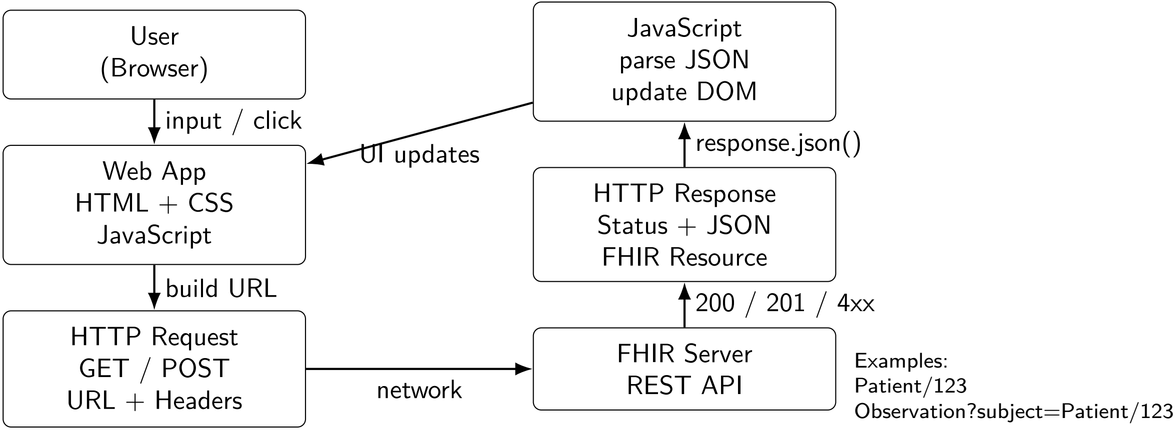
Today

- See how a web app talks to FHIR
- Understand why standards matter for AI
- Design your own FHIR-based mini solution

From Notebook to Web App

- Yesterday: direct HTTP requests via Python/Colab
- Today: simple web client (browser) calling FHIR server
- Same REST endpoints, different client

From Browser UI to FHIR Server (Architecture Overview)



- **Same mental model as Day 1:** URL + HTTP method + JSON
- **FHIR adds semantics:** resource types, references, codes (e.g., LOINC)
- **Tomorrow's focus:** turning REST calls into an interactive UI

Minimal JavaScript Example

```
const FHIR_SERVER = "https://example.org/fhir/";

async function loadPatient(patientId) {
        const response = await fetch(
        FHIR_SERVER + "Patient/" + patientId,
        { headers: { "Accept": "application/fhir+json" } }
        );
        const data = await response.json();
        console.log(data);
}
```

- Called from HTML (e.g., button click)
- Shows how UI, JavaScript, and FHIR connect

Simple UI Sketch (Conceptual)

- Input: Patient ID
- Button: "Load patient"
- Display: patient name, birthdate, list of Observations

Key Takeaways: Web Clients

- Browser acts as FHIR client
- REST endpoints remain the same
- Security (auth, HTTPS, tokens) is critical in real systems
- Today: we focus on data flow, not on full security model

Why AI Needs Good Data

- Machine learning models rely on:
  - large amounts of data
  - consistent structure
  - correct semantics
- Garbage in, garbage out:
  - inconsistent coding
  - missing values
  - ambiguous meaning
- Standards help address these issues

Role of Standards (FHIR, Terminologies)

- FHIR standardizes **structure** and **syntax**:
  - JSON layout
  - resource fields
  - references
- Terminologies standardize **meaning**:
  - LOINC for measurements
  - SNOMED CT for clinical concepts
  - ICD for diagnoses (administrative)
- Together: **semantic interoperability** $\rightarrow$ more reliable data for AI and analytics

Example: Vital Signs for Risk Prediction

- Data sources:
  - Observations of blood pressure, heart rate, temperature
- Standard coding:
  - LOINC codes for each measurement
- FHIR Observations provide:
  - numeric values
  - units
  - timestamps
  - patient reference
- AI-friendly dataset:
  - features: latest BP, average HR, etc.
  - labels: outcome (e.g., ICU admission, complication)

From FHIR to Feature Table (Conceptual)

- Step 1: extract Observations from FHIR server
- Step 2: filter by code and time
- Step 3: aggregate (mean, max, last value)
- Step 4: build a row per patient:
  - columns = features
- Step 5: feed into ML model

Toy Example: Simple Rule on FHIR Data

```json
{
        "resourceType": "Observation",
        "status": "final",
        "code": { ... },
        "subject": { "reference": "Patient/123" },
        "valueQuantity": { "value": 160, "unit": "mmHg" }
}
```

- Simple rule-based logic (not full AI!):
  - If systolic BP $>$ 140 mmHg $\rightarrow$ "high blood pressure alert"
- Demonstrates how structured FHIR data can drive simple decision rules

Trustworthy AI in Healthcare (High-Level)

- Beyond technical performance:
  - Transparency: Decisions and data sources must be understandable
  - Fairness: Performance should not systematically disadvantage patient groups
  - Robustness: Results must remain reliable under imperfect conditions
  - Accountability: Clear responsibility must exist for system outcomes
- Standards contribute to:
  - Traceability of input data
  - Consistent meaning across systems
  - Better documentation of data provenance
- Regulatory frameworks (e.g., EU AI Act) emphasize data quality and governance

Group Activity: FHIR-Based Mini Use Case

Task (in teams of 2–3)

- Choose a clinical scenario
- Identify FHIR resources needed
- Decide which REST operations are used
- Sketch the user interface
- Optionally add a simple alert rule

Example Scenario: Hypertension Monitoring

- Scenario:
  - Patient measures BP at home
  - Values are stored as Observations
  - Clinician receives a summary (DiagnosticReport)
- FHIR resources:
  - `Patient`, `Observation`, `DiagnosticReport`
- REST operations:
  - `POST /Observation`
  - `GET /Observation?subject=Patient/123`
- Optional alert:
  - If multiple Observations > 160 mmHg $\rightarrow$ flag as “urgent”

Group Output

- Use the template provided:
  - Clinical scenario (1–2 sentences)
  - FHIR resources involved
  - REST operations
  - UI sketch (very simple)
  - Optional AI/alert idea
- Prepare a 1–2 minute lightning presentation

Standards & AI References

- HL7 FHIR Official Spec: `https://www.hl7.org/fhir/`
- LOINC: `https://loinc.org/`
- SNOMED CT: `https://www.snomed.org/`
- Overview of AI in healthcare (example introductory sources, adapt as needed):
  - Topol, E. *Deep Medicine: How Artificial Intelligence Can Make Healthcare Human Again*.
  - Short tutorials and review articles from major journals (e.g., NEJM, Lancet Digital Health).

Wrap-Up

- FHIR and terminologies provide a solid foundation for interoperable data
- Web apps and services can use this foundation to build useful tools
- AI benefits from high-quality, standardised datasets
- You have:
  - practiced FHIR/REST in labs
  - seen a web client in action
  - designed your own mini use case

Thank you!